Relative Changes in Transcription Elongation, Splicing, and RNA Decay Kinetics from Total RNA-seq

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Abstract

Description of what I did.

1 Introduction

- RNA metabolism is a dynamic system, speed of reactions = kinetic rates. Kinetic rates are important (Furlan et al., 2021).
  - For example, Vock et al. (2025) recently estimated that variability in RNA degradation rate explains at least 25% of variability in transcript abundances between different human cell lines. In the immune system, control of RNA decay plays an especially important regulatory role (Schott and Stoecklin, 2010).
  - Transcription elongation speed also plays an important role in RNA metabolism. Changes in Pol II speed can affect several co-transcriptional processes, such as splicing, m^6A RNA methylation, and polyadenylation (Muniz et al., 2021). Recently, a study of Debès et al. (2023) showed that organism aging is associated with an increase in Pol II speed, whereas lifespan-prolonging interventions tend to decrease it.
  - The speed of intron splicing (time needed to splice the intron out of the transcript) is another important aspect of RNA metabolism. The estimates of intron splicing speed vary a lot, ranging from several seconds up to 15 minutes (Alpert et al., 2017; Shenasa and Bentley, 2023; Sethi et al., 2025).
  - I should also discuss transcription initiation rate, it's the main regulator of transcript abundance.
- Many other kinetic rates are important, e.g., Müller et al. (2024) recently showed that the speed of nuclear export is largely variable and influences the mRNA lifetime. Also, translation efficiency is widely studied (Gingold and Pilpel, 2011), as it affects protein abundance. However, our work here will focus only on rates of transcription initiation, elongation, splicing, and RNA degradation, as these rates affect total RNA-seq data and we will be able to capture them in our model.
- Measuring RNA kinetic rates is challenging, dedicated experiments are needed. Usually, these methods rely on RNA labeling (Furlan et al., 2021).

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  - Maybe I can elaborate a bit more on these methods, but only briefly, as they are not the focus of my work.

- Here, we discuss how total RNA-seq can be used to say something about these rates. Total RNA-seq data are common, easy to produce, and a large amount of them is already available. However, due to the 'static' nature of total RNA-seq data, our measurements will be a bit more limited.

1.1 Related Work

Here we briefly review methods most relevant to our approach; for a more comprehensive overview of computational methods for inferring RNA kinetic rates from sequencing data, we refer the reader to Furlan et al. (2021).

- Ameur et al. (2011) described the relation between intron coverage and transcription elongation speed in total RNA-seq. This principle was then used in subsequent works which attempted to model transcription elongation speed from total RNA-seq (Jonkers et al., 2014; Kawamura et al., 2019; Debès et al., 2023).

- The SnapSHot-Seq method (Gray et al., 2014) also uses the shape of intronic coverage, but assumes constant elongation speed, and uses the intronic coverage to compute other kinetic rates. Describes intron coverage as a mixture of unspliced and nascent introns exactly as I do.

- The INSPEcT- method (Furlan et al., 2020) is conceptually a bit similar to my work, but focuses on between-gene comparison rather than between-conditions comparison.

- The EISA method (Gaidatzis et al., 2015) is more similar to my approach, count intron and exon counts per gene and compute LFC of them between conditions.

- The REMBRANDTS method (Alkallas et al., 2017) builds on EISA, works with the assumption that splicing machinery may saturate for highly expressed genes, and somehow pools information across genes to remove this bias from the EISA-based method.

2 Dynamical Model of RNA Metabolism

We will now describe a simple dynamical model of RNA metabolism. For a more thorough description of the model and its assumptions, please see the section Supplementary Methods.

Mathematically, the model couples a transport partial differential equation (PDE) for the spatial density of elongating polymerases along the gene with a small set of ordinary differential equations (ODEs) for the abundances of completed (mature) transcripts and unspliced introns.

2.1 Model Schema and Kinetic Rates

Transcription is started by the RNA polymerase binding to the transcription start site (TSS) on DNA. Then, the RNA polymerase transcribes DNA by moving along it, elongating the nascent transcript. Once an intron is fully transcribed, the splicing machinery removes the intron from the transcript and degrades it. Finally, mature transcripts live in the cell until they are eventually degraded. The schema of this model is visualized in figure 1.

Our description focuses only on the processes in RNA metabolism that can be captured by total RNA-seq data. Other processes, such as nuclear export or translation, are not reflected in total RNA-seq, and we therefore omit them in our model.

The four processes described in our model - transcription initiation, transcription elongation, intron splicing and RNA degradation - are chemical reactions, and their speeds can be described by the following kinetic parameters:

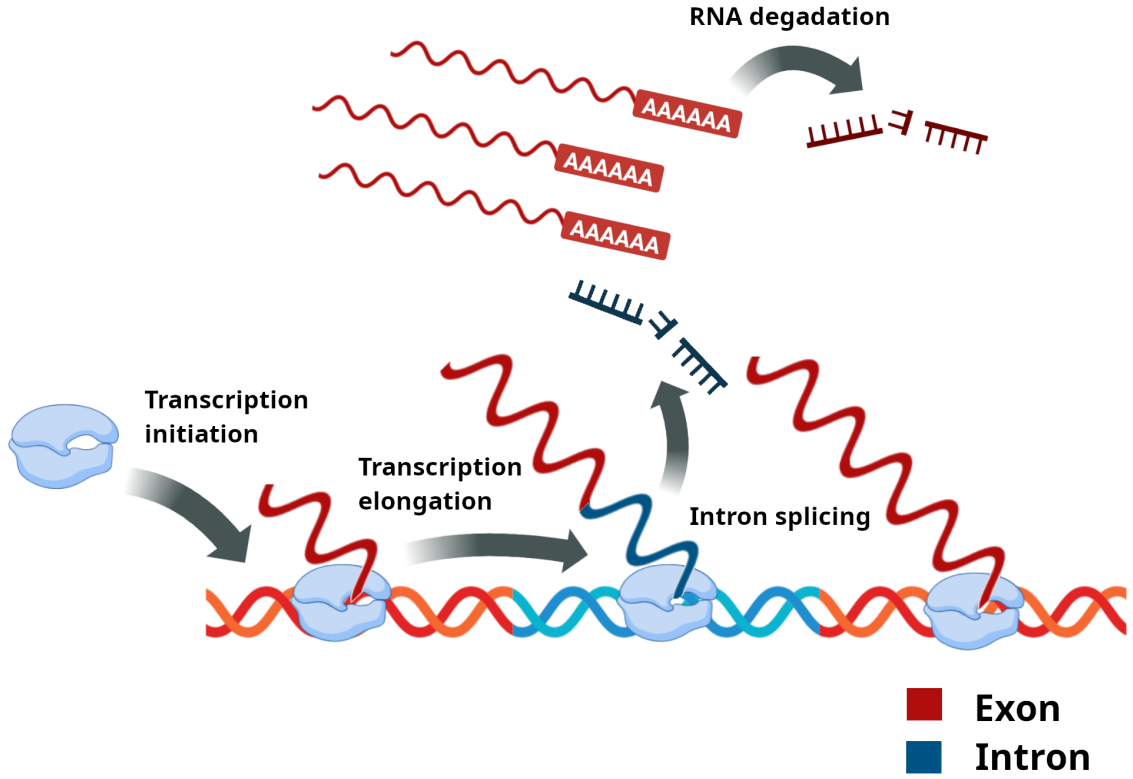


Figure 1: Schema of RNA metabolism.

- The transcription **initiation rate** k_{init}, measured by the number of polymerases that initiate transcription per second: $[k_{\text{init}}] = \text{polymerases} \cdot \text{s}^{-1}$.
- The transcription **elongation speed** k_{elong}, measured by the number of base-pairs transcribed per second: $[k_{\text{elong}}] = \text{bp} \cdot \text{s}^{-1}$.
- The intron **splicing speed** k_{splice}, measured as the inverse of the mean time required for intron splicing and degradation: $[k_{\text{splice}}] = \text{s}^{-1}$.
- The RNA **degradation rate** k_{deg}, measured as the inverse of the mean time that a mature transcript will live in a cell: $[k_{\text{deg}}] = \text{s}^{-1}$.

We assume that the kinetic rates may be different for different genes, and that the splicing speed may also vary for different introns in a given gene. Moreover, we assume that the kinetic rates may vary between conditions of the sample (e.g., comparing treated versus control samples), and this variation is what our statistical model (described in the section Inference Model) aims to estimate.

Besides these variations, we assume the kinetic rates to be constant. That is, while our model will describe how abundances of different RNA molecules change in time, the kinetic rates themselves are assumed to stay unchanged. Moreover, we assume that the elongation speed k_{elong} is constant across the whole gene; we will partially relax this assumption in the section Inference Model.

We also assume that transcription starts only at the TSS and terminates at the transcription end site (TES). That is, we assume no cryptic initiation within the gene body, and also no premature termination of transcription.

2.2 State Variables

We will describe the model state by the following variables: density of the polymerases transcribing the gene $\rho(x)$, number of mature transcripts T_{mature}, and, for each intron in the gene, the number of fully transcribed but unspliced introns $I_{\text{unspliced}}$. Furthermore, the polymerase density $\rho(x)$ can be used to derive the number of nascent transcripts T_{nascent} and the number of introns currently being transcribed I_{nascent}. These variables are described in more detail below.

Each of these variables is a function of time; we omit writing the explicit time argument in this section for the sake of brevity.

We assume a given gene (in a given biological sample) of a length L_{gene}. The gene length is measured in base-pairs, $[L_{\text{gene}}] = \text{bp}$. We consider the gene to be a continuous interval $[0, L_{\text{gene}}]$.

The polymerases transcribing the gene are modeled by a continuous density function $\rho(x), x \in [0, L_{\text{gene}}]$. The density is measured in the number of polymerases per base pair, $[\rho] = \text{polymerases} \cdot \text{bp}^{-1}$. The density $\rho(x)$ is non-negative and can be used to obtain the amount of polymerases located in any interval (a, b) as $\int_a^b \rho(x)dx$.

Specifically, we will denote the number of nascent transcripts (which is equal to the number of transcribing polymerases) as

$$T_{\text{nascent}} = \int_0^{L_{\text{gene}}} \rho(x)dx \tag{1}$$

Besides the number of nascent transcripts, we will also be interested in the number of polymerases that currently transcribe a given intron. We will consider an intron of length L_{intron}, starting at a location s. We denote the amount of nascent introns (i.e., the number of polymerases transcribing the intron) as

$$I_{\text{nascent}} = \int_s^{s+L_{\text{intron}}} \rho(x)dx \tag{2}$$

We denote the scalar amount of finished transcripts present in the sample as T_{mature}. We use the term *mature* loosely to mean a transcript whose transcription has finished; while most such transcripts are also fully processed, a fraction may still be undergoing processing (e.g. polyadenylation or splicing). We do not model these processing steps explicitly and treat all finished transcripts as a single pool T_{mature}.

Similarly, for a given intron, we denote the amount of finished but yet unspliced introns as $I_{\text{unspliced}}$. We don't make any assumptions about the transcript in which such an unspliced intron is located: the transcript may be nascent, transcribed but not processed yet, or the unspliced intron may be retained in an otherwise mature transcript.

2.3 Model Dynamics

We will now describe the dynamics of our model. For the sake of brevity, we will write the explicit time argument only for the variables that also depend on the location x.

To begin with, we define the polymerase flux as

$$J(x, t) = k_{\text{elong}} \cdot \rho(x, t) \tag{3}$$

The flux represents the rate at which polymerases pass the location x, and is measured in polymerases per second: $[J] = \text{polymerases} \cdot \text{s}^{-1}$.

We can now write the dynamics of our model:

Model dynamics

$$J(0,t) = k_{\text{init}} \tag{4a}$$

$$\frac{\partial \rho(x,t)}{\partial t} = -\frac{\partial J(x,t)}{\partial x} \tag{4b}$$

$$\frac{dT_{\text{mature}}}{dt} = J(L_{\text{gene}},t) - k_{\text{deg}} \cdot T_{\text{mature}} \tag{4c}$$

$$\frac{dI_{\text{unspliced}}}{dt} = J(s + L_{\text{intron}},t) - k_{\text{splice}} \cdot I_{\text{unspliced}} \tag{4d}$$

We will now discuss the model dynamics in more detail.

- Equation (4a) describes the **transcription initiation**. It is a boundary condition for the elongation dynamics (4b) at the start of the gene ($x = 0$). It states that polymerases enter the gene at $x = 0$ with rate k_{init}.
- Equation (4b) describes the **transcription elongation**. Polymerases move along the gene at speed k_{elong}, giving rise to the flux $J(x,t)$. Because polymerases are neither created nor destroyed within the gene body, the density at a given location changes only through the imbalance between the flux entering and leaving it — that is, the negative spatial gradient of the flux.
- Equation (4c) describes **transcript maturation** and **degradation**. The production term $J(L_{\text{gene}},t)$ is the flux of polymerases at the gene end, producing newly matured transcripts. The decay term $-k_{\text{deg}} \cdot T_{\text{mature}}$ models their degradation, treating T_{mature} as a well-mixed pool that decays exponentially.
- Equation (4d) describes **intron completion** and **splicing**. The production term $J(s + L_{\text{intron}},t)$ is the flux of polymerases at the intron end, producing newly completed introns. The decay term $-k_{\text{splice}} \cdot I_{\text{unspliced}}$ models intron splicing and degradation, again as exponential decay.

2.4 Steady-State Solution

We will now derive the steady-state solution of our model. Since total RNA-seq provides only a single snapshot in time, our inference model (described in the section Inference Model) assumes that the system is at steady state, i.e., the cells are in homeostasis.

To obtain the steady-state solution, we set time derivatives of all variables to zero. The full derivation is provided in the Supplementary Methods. We obtain the following solution:

Steady-state solution

$$\rho(x) = \frac{k_{\text{init}}}{k_{\text{elong}}} \tag{5a}$$

$$T_{\text{nascent}} = \frac{L_{\text{gene}} \cdot k_{\text{init}}}{k_{\text{elong}}} \tag{5b}$$

$$I_{\text{nascent}} = \frac{L_{\text{intron}} \cdot k_{\text{init}}}{k_{\text{elong}}} \tag{5c}$$

$$T_{\text{mature}} = \frac{k_{\text{init}}}{k_{\text{deg}}} \tag{5d}$$

$$I_{\text{unspliced}} = \frac{k_{\text{init}}}{k_{\text{splice}}} \tag{5e}$$

We will now comment on the steady-state solution described above.

- The density $\rho(x)$ is uniform over the whole gene body. Its value scales proportionately with k_{init} and inversely proportionally with k_{elong}.
- The values of T_{nascent} and I_{nascent} are obtained by integrating the density $\rho(x)$ over the gene and intron body, respectively. For this reason, they scale proportionally with L_{gene} and L_{intron}, respectively.
- In contrast, T_{mature} and $I_{\text{unspliced}}$ are pools of fully transcribed molecules, and therefore have no dependence on the gene or intron length. Each of them scales proportionally with the common production rate k_{init}, and inversely proportionally with their corresponding removal rate. The removal rate of mature transcripts is the degradation rate k_{deg}, while the removal rate of unspliced introns is k_{splice}, representing the speed of intron splicing and degradation.

2.5 Structural Non-identifiability

All steady-state quantities derived above depend on ratios of the kinetic rates, rather than on the individual rates directly. Consequently, observing these steady-state quantities is informative about ratios of the kinetic rates, while the individual rates remain non-identifiable.

This type of non-identifiability is common in dynamical models of RNA metabolism (Furlan et al., 2021), rather than being a special property of our model. We illustrate it here using a general ODE system; a more general treatment, which also covers the PDE formulation, is provided in the section Supplementary Methods. Consider an autonomous dynamical system with state vector $\boldsymbol{x}$ and dynamics

$$\frac{d\boldsymbol{x}}{dt} = f(\boldsymbol{x}). \tag{6}$$

For any $c \in \mathbb{R}_+$, it holds that

$$f(\boldsymbol{x}) = 0 \Leftrightarrow c \cdot f(\boldsymbol{x}) = 0. \tag{7}$$

The steady states of the system are defined as the points where $f(\boldsymbol{x}) = 0$; therefore, steady-state observations cannot distinguish the dynamics $f(\boldsymbol{x})$ from $c \cdot f(\boldsymbol{x})$. In a model of metabolism, $c \cdot f(\boldsymbol{x})$ describes a system in which all reactions proceed c times faster than in the system with dynamics $f(\boldsymbol{x})$; both systems share the same steady states, and thus cannot be told apart from steady-state data alone. Because scaling all rates by a common factor leaves the observations

unchanged, any identifiable quantity must be unchanged by such scaling, which is exactly the case for ratios of rates.

Because only ratios of rates are identifiable, comparing biological conditions can reveal changes only in these ratios. As the logarithm of a ratio is a difference of logarithms, the log fold change (LFC) of a rate ratio equals the difference of the individual rate LFCs, for example $\mathrm{LFC}(k_{\mathrm{elong}}/k_{\mathrm{deg}}) = \mathrm{LFC}(k_{\mathrm{elong}}) - \mathrm{LFC}(k_{\mathrm{deg}})$. Such differences of LFCs are therefore the quantities that our inference model estimates.

Finally, we note that the compositional nature of RNA-seq data introduces a second, independent source of non-identifiability. We describe it, and which of the estimated quantities it affects, in the section Compositional Non-identifiability.

3 Inference Model

Here we describe our inference model, working with total RNA-seq data. As shown in the section Structural Non-identifiability, the individual kinetic rates are not identifiable from steady-state observations, and only their ratios can be estimated. Our inference model focuses on comparing biological conditions, and estimates LFCs of these ratios between the conditions. Throughout this section, we assume the system is at steady state, and, unless stated otherwise, we retain the modeling assumptions introduced in the section Dynamical Model of RNA Metabolism.

3.1 Distinguishing Nascent and Unspliced Introns by Read Coverage

We will start by discussing the key component of our model: the fact that nascent introns I_{nascent} and unspliced introns $I_{\mathrm{unspliced}}$ produce different read coverage patterns, which allows us to distinguish them, and consequently identify the ratio $k_{\mathrm{elong}}/k_{\mathrm{splice}}$ from the data. This core principle was first observed by Ameur et al. (2011), and subsequently utilized in the work of others (Jonkers et al., 2014; Gray et al., 2014; Kawamura et al., 2019; Debès et al., 2023).

The read coverage can be decomposed into two parts: the location of reads (the 'shape' of coverage), and the amount of reads (the 'scale' of coverage). Here we focus on the shape; the scale enters our model through the read counts, as described below.

We now consider an intron of length L_{intron}; without loss of generality, we assume that it starts at position $s = 0$. We will model each short sequencing read as a point $x \in [0, L_{\mathrm{intron}}]$. The shape of read coverage can then be described by a probability density $p(x)$, specifying the location of reads. Nascent and unspliced introns produce reads with different location distributions; the overall density $p(x)$ is therefore a mixture distribution:

$$p(x) = \pi \cdot p_{\mathrm{nascent}}(x) + (1 - \pi) \cdot p_{\mathrm{unspliced}}(x) \tag{8}$$

where p_{nascent} is the location distribution of reads from nascent introns, $p_{\mathrm{unspliced}}$ is the location distribution of reads from unspliced introns, and π is a mixture coefficient giving the proportion of reads from nascent introns:

$$\pi = \frac{I_{\mathrm{nascent}}}{I_{\mathrm{nascent}} + 2 \cdot I_{\mathrm{unspliced}}} \tag{9}$$

The constant term 2 accounts for the fact that a nascent intron is, on average, only half transcribed, and therefore produces about half as many reads as a fully transcribed unspliced intron. This follows from the uniform polymerase density (5a).

Plugging in the steady-state solution for I_{nascent} and $I_{\mathrm{unspliced}}$, we obtain:

$$\pi = \frac{L_{\mathrm{intron}} \cdot k_{\mathrm{splice}}}{L_{\mathrm{intron}} \cdot k_{\mathrm{splice}} + 2 \cdot k_{\mathrm{elong}}} \tag{10}$$

We also note that $L_{\text{intron}} \cdot k_{\text{elong}}^{-1}$ is the time needed to transcribe the intron, and k_{splice}^{-1} is the time needed to splice it out. Using that, we may also write

$$\pi = \frac{\text{transcription time}}{\text{transcription time} + 2 \cdot \text{splicing time}} \tag{11}$$

We will now describe the mixture components $p_{\text{nascent}}(x)$ and $p_{\text{unspliced}}(x)$. We begin with $p_{\text{unspliced}}(x)$: these reads come from fully transcribed introns, and we simply model their location by a uniform distribution

$$p_{\text{unspliced}}(x) = \frac{1}{L_{\text{intron}}} \tag{12}$$

On the other hand, $p_{\text{nascent}}(x)$ is produced by nascent introns, which have not finished transcribing the intron interval yet. Due to this, reads at location x can be produced only from nascent introns that are already elongated past x. This gives us:

$$p_{\text{nascent}}(x) \propto \int_x^{L_{\text{intron}}} \rho(z) dz \tag{13}$$

The steady-state solution (5a) gives a polymerase density $\rho(x)$ that is constant over the gene body. Carrying out the integral in (13) and normalizing to a probability density gives us:

$$p_{\text{nascent}}(x) = \frac{2L_{\text{intron}} - 2x}{L_{\text{intron}}^2} \tag{14}$$

The two mixture components thus have distinctly different shapes: a linearly decreasing density for nascent introns, and a uniform density for unspliced introns. The shape of the observed intron coverage is therefore informative about the mixture coefficient π, and, through (10), about the ratio $k_{\text{elong}}/k_{\text{splice}}$.

An illustration of the described coverage pattern is presented in the figure 2, using a total RNA-seq dataset of the K562 human cell line. The decrease in coverage along the intron is clearly visible when the coverage is aggregated over many introns, while the coverage of a single intron is much noisier.

We note that the same argument can also be applied to whole transcripts: the fraction of exonic reads originating from nascent transcripts is

$$\begin{aligned} \pi_{\text{gene}} &= \frac{T_{\text{nascent}}}{T_{\text{nascent}} + 2 \cdot T_{\text{mature}}} = \frac{L_{\text{gene}} \cdot k_{\text{deg}}}{L_{\text{gene}} \cdot k_{\text{deg}} + 2 \cdot k_{\text{elong}}} \\ &= \frac{\text{gene transcription time}}{\text{gene transcription time} + 2 \cdot \text{mean transcript lifetime}} \end{aligned} \tag{15}$$

The mature transcripts produce reads with a uniform distribution of their location, while nascent transcripts produce reads with coverage decreasing over the gene body, analogously to the case of intron coverage we described above. However, exons may also be spliced out in alternative splicing, which affects the resulting exon coverage and complicates its use for inference.

For most genes, the transcription time is much shorter than the mean transcript lifetime, and only a few percent of exonic reads therefore originate from nascent transcripts. In our inference model, we neglect the nascent transcript fraction, and assume that all exonic reads originate from mature transcripts. A further quantification of the effect on our inference model is provided in the section Supplementary Methods.

Finally, read coverage is also affected by technical factors, such as read mappability or GC content (Derrien et al., 2012; Love et al., 2016). The observed coverage is therefore distorted in a sequence-dependent manner, so that the location distributions derived above are only approximate. As such distortions are determined by the underlying sequence, they affect all samples in the same way; we discuss their impact on our estimates in the section Supplementary Methods.

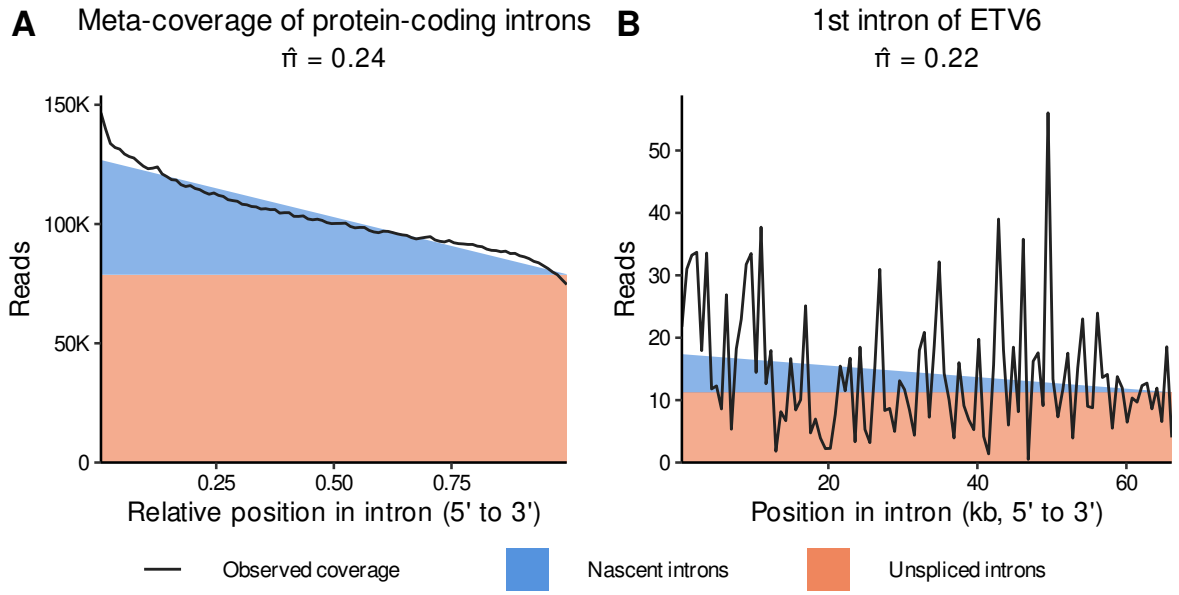


Figure 2: Read coverage of introns in K562 cells, using a total RNA-seq dataset of The ENCODE Project Consortium (2012) (accession ENCSR000AEL, two replicates pooled). The decrease in coverage along the intron is clearly visible in the meta-coverage, while the coverage of a single intron is much noisier. In both panels, the black line is the observed coverage, and the shaded areas are the fitted mixture (8) of nascent introns (blue, linearly decreasing density (14)) and unspliced introns (orange, uniform density (12)); $\hat{\pi}$ is the estimated fraction of reads originating from nascent introns. **(A)** Meta-coverage of all 166,239 protein-coding introns, containing 10.3 million reads in total. **(B)** Coverage of a single intron, the first intron of the *ETV6* gene (66,426 bp, 1,432 reads). The coverage of every intron is rescaled to 100 equally sized bins, and normalized so that the binned values sum to the number of reads of that intron; one bin in panel B therefore spans 664 bp, and features narrower than a bin appear smoothed. The mixture is fitted here by maximum likelihood to the coverage shown. Our inference model also uses maximum likelihood, but parametrizes the mixture differently: there, π is not an independent parameter, but a function of several other parameters, which are all estimated jointly (see the section Inference Model), and it is always obtained for individual introns, rather than for a meta-coverage. The figure is therefore only an illustration of the coverage pattern, and does not show the output of our inference model.

3.2 Model Parametrization

We will now describe the parametrization of our model. The goal of our inference task is to estimate the changes in ratios of kinetic rates between biological conditions. The model is fitted on total RNA-seq data using maximum likelihood estimation, as described below. Throughout the paper, we write vectors and matrices in bold, and scalars in plain type.

We assume a dataset consisting of n samples. The samples are annotated by a design matrix $\boldsymbol{X} \in \mathbb{R}^{n \times q}$, where q is the number of features. The features may be, for example, indicator columns of categorical annotations, or continuous variables. The intercept column is not supposed to be a part of the design matrix, as our model handles intercept terms as explicit parameters. The intercept terms therefore correspond to the reference condition, that is, to samples whose row of $\boldsymbol{X}$ is zero.

We will now describe a model fitted on a specific gene, with $m \geq 1$ introns. We will discuss variants of this model, including one fitted on data from multiple genes jointly later on.

Our inference model relaxes one assumption of the section Dynamical Model of RNA Metabolism: we require the elongation speed to be constant only within each intron, and denote by $k_{\text{elong},i}$ the elongation speed over the intron i. The polymerase density then remains constant within each intron, equal to $k_{\text{init}}/k_{\text{elong},i}$, so the results (5c), (10) and (14) hold as derived, with k_{elong} replaced by $k_{\text{elong},i}$. The splicing speed may differ between the introns of a gene, as assumed in the section Dynamical Model of RNA Metabolism, and we denote by $k_{\text{splice},i}$ the splicing speed of the intron i. We likewise denote by $L_{\text{intron},i}$ the length of the intron i.

Our dynamical model has four kinetic rates, and as described in the section Structural Non-identifiability above, only ratios of these rates are identifiable. There are $\binom{4}{2} = 6$ unordered pairs of rates, and thus six such ratios up to inversion. However, their LFCs span only a three-dimensional space, and our model thus parametrizes only three of them. The choice of which LFCs to parametrize is only a matter of convention; we choose the three ratios that share k_{deg} as their denominator. We write each LFC as a vector of coefficients with respect to the design matrix, so that $\text{LFC}(\cdot) \in \mathbb{R}^q$, and the LFC in the individual samples is obtained as $\boldsymbol{X}\,\text{LFC}(\cdot) \in \mathbb{R}^n$. We denote the three LFCs as

$$\boldsymbol{\alpha} = \text{LFC}(k_{\text{init}}/k_{\text{deg}}) \tag{16a}$$

$$\boldsymbol{\beta}_i = \text{LFC}(k_{\text{elong},i}/k_{\text{deg}}) \tag{16b}$$

$$\boldsymbol{\gamma}_i = \text{LFC}(k_{\text{splice},i}/k_{\text{deg}}) \tag{16c}$$

The initiation and degradation rates are properties of the whole gene, giving us a single $\boldsymbol{\alpha}$, while the elongation and splicing speeds may differ between the introns, so we have one $\boldsymbol{\beta}_i$ and one $\boldsymbol{\gamma}_i$ for each of the m introns. Each of these parameters is a vector in $\mathbb{R}^q$, holding one coefficient per feature of the design matrix.

The remaining three LFCs are obtained as their differences, for each intron i,

$$\text{LFC}(k_{\text{init}}/k_{\text{elong},i}) = \boldsymbol{\alpha} - \boldsymbol{\beta}_i \tag{17a}$$

$$\text{LFC}(k_{\text{init}}/k_{\text{splice},i}) = \boldsymbol{\alpha} - \boldsymbol{\gamma}_i \tag{17b}$$

$$\text{LFC}(k_{\text{elong},i}/k_{\text{splice},i}) = \boldsymbol{\beta}_i - \boldsymbol{\gamma}_i \tag{17c}$$

Reversing the order of the rates in a ratio flips the sign of its LFC. All six ratio LFCs are therefore determined by $\boldsymbol{\alpha}$, $\boldsymbol{\beta}_i$ and $\boldsymbol{\gamma}_i$.

This parametrization corresponds directly to the steady-state solution derived in the section Steady-State Solution. The mature transcript pool (5d) changes by $\text{LFC}(T_{\text{mature}}) = \boldsymbol{\alpha}$, which is the quantity reported by a standard differential expression analysis; the nascent pool of the intron i (5c) changes by $\boldsymbol{\alpha} - \boldsymbol{\beta}_i$, and its unspliced pool (5e) by $\boldsymbol{\alpha} - \boldsymbol{\gamma}_i$, following (17a) and

(17b). The remaining combination $\boldsymbol{\beta}_i - \boldsymbol{\gamma}_i$ (17c) is the LFC of the ratio $k_{\text{elong},i}/k_{\text{splice},i}$, which determines the coverage shape of the intron i through (10).

The model predicts several quantities, which give rise to the likelihood function described in the section Model Likelihood. Each of the predicted quantities is a vector in $\mathbb{R}^n$, holding one value per sample:

- The number of exonic reads $\boldsymbol{\mu}^{\text{exon}}$.
- The number of intronic reads $\boldsymbol{\mu}_i^{\text{intron}}$ for each intron i.
- The mixture coefficient $\boldsymbol{\pi}_i$ of the mixture (8), specifying the shape of the coverage of the intron i.

The predictions also utilize several nuisance parameters:

- Intercept term for exon read count c^{exon}.
- Intercept term for intron read count c_i^{intron}, specified for each intron i.
- Intercept term c_i^{π} for each intron i, which will be used for prediction of the intron coverage parameter $\boldsymbol{\pi}_i$.

Besides these nuisance parameters, we also utilize the following offset terms, estimated independently of the model parameters. Both are vectors in $\mathbb{R}^n$, holding one value per sample:

- The library size $\boldsymbol{\ell}$, reflecting the sequencing depth. The same $\boldsymbol{\ell}$ is used for every gene.
- The average transcript length $\boldsymbol{L}^{\text{tx}}$, reflecting the effect of alternative exon splicing. It is estimated for each gene separately.

The model predictions are then:

Model predictions

$$\boldsymbol{\mu}^{\text{exon}} = \exp\left(\boldsymbol{X}\boldsymbol{\alpha} + c^{\text{exon}} + \log\boldsymbol{\ell} + \log\boldsymbol{L}^{\text{tx}}\right) \tag{18a}$$

$$\boldsymbol{\mu}_i^{\text{nascent}} = \exp\left(\boldsymbol{X}(\boldsymbol{\alpha} - \boldsymbol{\beta}_i) + c_i^{\text{intron}} + c_i^{\pi} + \log\boldsymbol{\ell}\right) \tag{18b}$$

$$\boldsymbol{\mu}_i^{\text{unspliced}} = \exp\left(\boldsymbol{X}(\boldsymbol{\alpha} - \boldsymbol{\gamma}_i) + c_i^{\text{intron}} + \log\boldsymbol{\ell}\right) \tag{18c}$$

$$\boldsymbol{\mu}_i^{\text{intron}} = \boldsymbol{\mu}_i^{\text{nascent}} + \boldsymbol{\mu}_i^{\text{unspliced}} \tag{18d}$$

$$\boldsymbol{\pi}_i = \boldsymbol{\mu}_i^{\text{nascent}} / \boldsymbol{\mu}_i^{\text{intron}} \tag{18e}$$

We will now comment on the model predictions described above.

- The division in (18e) is applied elementwise.
- The predicted intronic reads are split into the two pools of the mixture (8): reads from nascent introns (18b), and reads from unspliced introns (18c). Their changes between conditions are $\boldsymbol{\alpha} - \boldsymbol{\beta}_i$ and $\boldsymbol{\alpha} - \boldsymbol{\gamma}_i$, matching the steady-state pools (5c) and (5e). The two pools are only intermediate quantities, and enter the likelihood through $\boldsymbol{\mu}_i^{\text{intron}}$ and $\boldsymbol{\pi}_i$.
- The prediction of $\boldsymbol{\pi}_i$ (18e) is the fraction of intronic reads originating from nascent introns, as defined in (9). Dividing through by $\boldsymbol{\mu}_i^{\text{nascent}}$ gives the equivalent form $\boldsymbol{\pi}_i = \sigma\left(c_i^{\pi} + \boldsymbol{X}(\boldsymbol{\gamma}_i - \boldsymbol{\beta}_i)\right)$, where $\sigma(x) = \exp(x)/(1+\exp(x))$ is the sigmoid function.

- The intercept c_i^π therefore absorbs all factors that shift the coverage shape of the intron i but do not change between the conditions. Following (10), it equals $\log(L_{\text{intron},i}/2) + \log(k_{\text{splice},i}/k_{\text{elong},i})$, with the rates taken in the reference condition. The intron lengths are for this reason not needed as an input to our model.

3.3 Model Likelihood

We will now describe the likelihood function of our model, which we use for parameter estimation by the maximum likelihood method, and as the basis of the statistical tests. It has one term for each of the quantities predicted in the section Model Parametrization: the exonic reads, the intronic reads, and the shape of the intron coverage.

The observed data consist of the exonic and intronic read counts, and of the distribution of the intronic reads along the intron. The exonic and intronic reads are disjoint sets, that is, no individual read is counted as both exonic and intronic. To describe the distribution of the reads along an intron, we divide each intron into b bins of equal width, and express the positions relative to the intron length, on the interval $[0, 1]$.

The model is fitted to the following observed data:

- The exon read counts $\boldsymbol{y}^{\text{exon}} \in \mathbb{N}_0^n$. The reads are aggregated over all exons of the gene, for example by aligning them to a reference transcriptome.
- The intron read counts $\boldsymbol{y}_i^{\text{intron}} \in \mathbb{N}_0^n$ for each intron i.
- The intron coverage $\boldsymbol{Y}_i^{\text{cov}} \in \mathbb{R}^{n \times b}$ for each intron i, holding the number of reads falling into each bin, for every sample.

The read counts are vectors holding one value per sample, and correspond to the predictions (18a) and (18d). The coverage is a matrix, holding one row per sample and one column per bin, and corresponds to the prediction (18e).

We treat every read as a single point on its intron, and do not model the shape of its alignment. An alignment spans a region of the genome, and need not even be contiguous, for example when the mates of a read pair are separated by a gap. The introns are typically long compared to the reads, so we consider this a minor simplification; our implementation rescales the per-base coverage of the intron into $b = 100$ bins. Other values may be used as well: setting b to the length of the intron, separately for each intron, gives one bin per base, so the unbinned coverage is a special case of the description above.

The coverage of each sample sums to the read count of its intron,

$$\boldsymbol{y}_i^{\text{intron}} = \boldsymbol{Y}_i^{\text{cov}} \mathbf{1} \tag{19}$$

where $\mathbf{1} \in \mathbb{R}^b$ is a vector of ones. The intron read counts and the intron coverage are therefore not two independent observations: the counts are determined by the coverage. We will factor the likelihood of the intron i as

$$P(\boldsymbol{Y}_i^{\text{cov}}) = P(\boldsymbol{y}_i^{\text{intron}})\, P(\boldsymbol{Y}_i^{\text{cov}} \mid \boldsymbol{y}_i^{\text{intron}}) \tag{20}$$

where the first factor describes the number of intronic reads, and the second one their positions within the intron.

The exonic and intronic reads, on the other hand, are disjoint by construction. We therefore treat the exonic count and the counts of the individual introns as independent observations, so that the likelihood of a gene with m introns factorizes, using (20), as

$$P\left(\boldsymbol{y}^{\text{exon}}, \boldsymbol{Y}_1^{\text{cov}}, \ldots, \boldsymbol{Y}_m^{\text{cov}}\right) = P(\boldsymbol{y}^{\text{exon}}) \prod_{i=1}^{m} P(\boldsymbol{y}_i^{\text{intron}})\, P(\boldsymbol{Y}_i^{\text{cov}} \mid \boldsymbol{y}_i^{\text{intron}}) \tag{21}$$

The three kinds of factors give the three terms of the log-likelihood, which we describe next.

We index the samples by $s = 1, \ldots, n$ and the coverage bins by $j = 1, \ldots, b$, keeping the index $i = 1, \ldots, m$ of the introns. We write y_s^{exon} for the element of $\boldsymbol{y}^{\text{exon}}$ belonging to the sample s, and use the same convention for the other observed and predicted quantities, so that $Y_{i,s,j}^{\text{cov}}$ is the number of reads of the sample s falling into the bin j of the intron i.

We model the read counts as Poisson-distributed, with the means given by the predictions (18a) and (18d),

$$y_s^{\text{exon}} \sim \text{Poisson}\left(\mu_s^{\text{exon}}\right) \tag{22a}$$

$$y_{i,s}^{\text{intron}} \sim \text{Poisson}\left(\mu_{i,s}^{\text{intron}}\right) \tag{22b}$$

independently over the samples, and, within a sample, independently over the exonic counts and the counts of the individual introns.

The first two factors of (21) therefore give the standard Poisson log-likelihood, summed over the samples and, in the case of the introns, over the introns of the gene. We denote these two terms by $\mathcal{L}^{\text{exon}}$ and $\mathcal{L}^{\text{intron}}$. The third term, describing the positions of the intronic reads, is derived below.

In the section Distinguishing Nascent and Unspliced Introns by Read Coverage, we described the read locations by a coordinate $x \in [0, L_{\text{intron}}]$, measured from the start of the intron. We now express the locations relative to the intron length, so that $x \in [0, 1]$ holds for every intron, with $x = 0$ at its start and $x = 1$ at its end. We denote by x_j the center of the bin j, and take it as the location of all reads falling into that bin.

Expressing the positions relative to the intron length amounts to setting $L_{\text{intron},i} = 1$ in the densities (12) and (14). The mixture (8) of the intron i in the sample s is therefore

$$p_{i,s}(x) = \pi_{i,s} \cdot 2(1 - x) + (1 - \pi_{i,s}) = 1 + \pi_{i,s}(1 - 2x) \tag{23}$$

The intron length cancels out, which is the same observation we made about the intercept c_i^{π} in the section Model Parametrization: the intron lengths are not needed as an input to our model.

The binning of the intron amounts to a rounding of the read locations, as we treat every read falling into the bin j as located at its center x_j. Conditionally on the number of intronic reads, the numbers of reads in the individual bins then follow a multinomial distribution, whose probabilities are obtained by evaluating the density (23) at the bin centers and normalizing,

$$p_{i,s,j}^{\text{bin}} = \frac{p_{i,s}(x_j)}{\sum_{k=1}^{b} p_{i,s}(x_k)} = \frac{1}{b}\left(1 + \pi_{i,s}(1 - 2x_j)\right) \tag{24}$$

The second equality follows from the bin centers being symmetric around 1/2, so that $\sum_k (1 - 2x_k) = 0$.

The third term of the log-likelihood is then

$$\mathcal{L}^{\text{cov}} = \sum_{i=1}^{m} \sum_{s=1}^{n} \sum_{j=1}^{b} Y_{i,s,j}^{\text{cov}} \log\left(1 + \pi_{i,s}(1 - 2x_j)\right) \tag{25}$$

where we dropped the multinomial coefficients and the normalizing factor $1/b$ of (24), which are just additive constants of the log-likelihood.

The log-likelihood of the gene is the sum of the three terms,

$$\mathcal{L} = \mathcal{L}^{\text{exon}} + \mathcal{L}^{\text{intron}} + \mathcal{L}^{\text{cov}} \tag{26}$$

which we use for parameter estimation and statistical inference, as described below.

3.4 Estimation and Inference

We estimate the parameters of our model by the maximum likelihood method, taking as the estimate the values of the parameters that maximize the log-likelihood (26). The estimated parameters are the LFCs $\boldsymbol{\alpha}$, $\boldsymbol{\beta}_i$ and $\boldsymbol{\gamma}_i$, and the intercepts c^{exon}, c_i^{intron} and c_i^{π}. The parameters of primary interest are $\boldsymbol{\beta}_i$ and $\boldsymbol{\gamma}_i$. The intercepts are nuisance parameters, and $\boldsymbol{\alpha}$ is the change in the mature transcript pool, which is the quantity reported by a standard differential expression analysis.

The offset terms $\boldsymbol{\ell}$ and $\boldsymbol{L}^{\text{tx}}$ are not estimated by our model. We obtain them in the same way as a standard differential expression pipeline does: the library sizes by the median-of-ratios method (Anders and Huber, 2010; Love et al., 2014), and the average transcript lengths from a transcript-level quantification of each sample (Soneson et al., 2016; Patro et al., 2017). Both are estimated before the model is fitted and are held fixed during the fitting.

The log-likelihood (26) has no maximum in a closed form, and we therefore maximize it numerically. It is differentiable in all model parameters, so both the gradients used by the optimizer and the second derivatives used for the Fisher information matrix below are obtained by automatic differentiation (Ansel et al., 2024). The log-likelihood is not concave, and the optimization is for this reason not guaranteed to reach the global maximum. A suboptimal solution affects not only the estimates, but also the test statistics described below.

The maximum of the log-likelihood also need not be attained at a finite value of the parameters. The coverage term (25) decomposes over the introns and the samples, and its contribution from a given intron and sample is a concave function of $\pi_{i,s}$ on the interval $[0, 1]$, whose maximum may lie at one of the corners of that interval. The corner $\pi_{i,s} = 0$ is optimal when the observed coverage shows no bias towards the start of the intron, and the corner $\pi_{i,s} = 1$ when the coverage falls off at least as steeply as the nascent density $2(1 - x)$. The mixture coefficient is obtained through the sigmoid function, $\boldsymbol{\pi}_i = \sigma(c_i^{\pi} + \boldsymbol{X}(\boldsymbol{\gamma}_i - \boldsymbol{\beta}_i))$, which is the equivalent form of the prediction (18e), as we noted in the section Model Parametrization. A corner solution is therefore approached only as the argument of the sigmoid goes to $-\infty$ or $+\infty$, so the parameters entering it, that is $\boldsymbol{\beta}_i$, $\boldsymbol{\gamma}_i$ and c_i^{π}, may diverge. Whether they do is a property of the whole dataset rather than of a single sample: the samples share these parameters, and the design matrix constrains how $\boldsymbol{\pi}_i$ may differ between them, so a corner preferred by one sample may be ruled out by the coverage observed in the others. The other terms of the likelihood do not prevent the divergence, as the predicted counts (18d) stay finite while $\boldsymbol{\pi}_i$ saturates.

When the divergence does occur, the log-likelihood is flat along the diverging direction. The parameter combination corresponding to such a direction is not identifiable, and the Fisher information matrix is rank deficient in that direction. Only that direction is affected, and the other parameters may still be estimated well; a fit may, for example, diverge in c_i^{π} and $\boldsymbol{\beta}_i$ while $\boldsymbol{\gamma}_i$ stays well behaved. The optimizer stops at a large but finite value in practice, so the problem shows up as an ill-conditioned Fisher information matrix rather than an exactly singular one. A similar difficulty is known in logistic regression as the separation of data, where the maximum likelihood estimate does not exist whenever the linear predictor can separate the two classes perfectly (Albert and Anderson, 1984).

The corner solutions do not arise only by chance, that is, from the sampling of the reads. Sequence-dependent technical biases may lead to them as well, for example when the GC content influences the number of reads produced per location. An incorrect annotation of the gene, such as an unannotated transcription start site within the intron, acts in the same way. The mixture (23) is then not a correct description of the read locations, and the observed coverage need not resemble any shape it can produce.

We may test the individual parameters of the model by the Wald test, which is inexpensive to compute, as it needs only the observed Fisher information matrix, that is, the matrix of the second derivatives of the negative log-likelihood at the estimate. Its inverse estimates the covariance matrix of the parameters, giving their standard errors, and each parameter is then

tested against zero by comparing its estimate divided by the standard error to the standard normal distribution.

The Fisher information matrix is, however, not invertible when the log-likelihood is flat in some direction, so we do not invert it directly. We instead obtain its rank, and use a QR decomposition with column pivoting to select a subset of the parameters whose submatrix has full rank. The standard errors are computed from that submatrix, while the remaining parameters are reported as non-identifiable, without a standard error or a p-value. The pivoting therefore provides an identifiability flag for each parameter, which we report together with the estimates.

Moreover, the Wald test relies on a quadratic approximation of the log-likelihood near the estimate, which may be poor when the log-likelihood is skewed or nearly flat, and its results may then differ substantially from those of the likelihood ratio test described below (Hauck and Donner, 1977). Together with the fact that the Wald test leaves some of the parameters untested, this leads us to prefer the likelihood ratio test, which we use as the default test of our implementation. We nevertheless compute the Wald test as well, both for the identifiability flag and for the standard errors, which we use when setting the strength of the regularization described below.

The likelihood ratio test compares the fitted model with a null model, obtained by imposing a linear restriction on its parameters. In our implementation, the restriction sets the tested LFC to zero, and we impose it by a reduced design matrix, obtained by removing the tested feature from the design matrix, so that the two models are nested. Both models are fitted by the procedure described above, and we denote their maximized log-likelihoods by $\mathcal{L}_{\text{full}}$ and $\mathcal{L}_{\text{reduced}}$. The test statistic

$$\Lambda = 2\left(\mathcal{L}_{\text{full}} - \mathcal{L}_{\text{reduced}}\right) \tag{27}$$

follows the χ^2 distribution asymptotically under the null hypothesis, with the degrees of freedom given by the difference between the numbers of the free parameters of the two models. We obtain the p-value of the test from that distribution, and adjust the resulting p-values for multiple testing. The test statistic requires only the two maximized log-likelihoods, and not the Fisher information matrix, so the flat directions of the log-likelihood do not prevent us from computing it.

The flat directions of the log-likelihood may drive the estimates to extreme values, and a common remedy is to regularize the model by penalizing large values of its parameters. We use an L2 regularization of the LFCs that these directions affect, and maximize

$$\mathcal{L}^{\text{reg}} = \mathcal{L} - \sum_{i=1}^{m}\sum_{k=1}^{q}\left(\lambda_k^{\beta}\beta_{i,k}^2 + \lambda_k^{\gamma}\gamma_{i,k}^2\right) \tag{28}$$

where $\beta_{i,k}$ and $\gamma_{i,k}$ are the coefficients of $\boldsymbol{\beta}_i$ and $\boldsymbol{\gamma}_i$ belonging to the feature k of the design matrix, and λ_k^{β} and λ_k^{γ} are the corresponding regularization strengths. Maximizing (28) gives the maximum a posteriori estimate under a normal prior with mean zero and variance $\sigma^2 = 1/(2\lambda)$, placed on each of the regularized coefficients.

We set the strength of the regularization from the data, following the same principle as the shrinkage of the LFCs in a differential expression analysis (Love et al., 2014). We first fit every gene without the regularization, and collect the estimates of a given parameter and feature across all the genes, together with their standard errors from the Wald test. We fit a prior distribution to these estimates by the adaptive shrinkage method (Stephens, 2016), and take its standard deviation as σ, which gives the regularization strength $\lambda = 1/(2\sigma^2)$. Only the estimates that are identifiable and have a finite standard error enter this step. Each gene is then fitted again with the regularization. The regularization strengths are therefore shared by all the genes, while being specified separately for each parameter and each feature of the design matrix.

Of the parameters whose estimates may diverge, we regularize only $\boldsymbol{\beta}_i$ and $\boldsymbol{\gamma}_i$, and leave the intercept c_i^{π} unregularized. Zero is a meaningful value to shrink the LFCs towards, as it

corresponds to no change between the conditions, while c_i^π describes the reference condition itself. The regularization changes the reported estimates, but not the p-values, which are obtained from the unregularized fit.

The parameters are fitted jointly, so all of them may move to compensate for the shrinkage, and even a regularized estimate may grow in magnitude as a result. The strength of the regularization depends on the standard errors of the Wald test, and inherits their limitations. The regularization also does not address the model misspecification discussed above, as it shrinks the estimates it produces in the same way as any other.

A different way of pooling the information across the genes is to fit a single LFC shared by all of them, which is done by the global model described in the section Model Variants.

3.5 Model Variants

The model described above is the intron-specific variant, with a separate $\boldsymbol{\beta}_i$ and $\boldsymbol{\gamma}_i$ for every intron of a gene. These parameters, together with the intercept c_i^π, belong to a single intron, and the reads of that intron are the only data specific to it. The statistical power of such an estimate is limited, and the estimate is also sensitive to technical biases affecting the coverage of that one intron. Sharing $\boldsymbol{\beta}_i$ and $\boldsymbol{\gamma}_i$ across several introns, or across several genes, reduces the influence of any single intron, at the price of the granularity of the results. The intercepts are never shared, as c_i^π absorbs the length of the intron and the ratio of its kinetic rates in the reference condition, which differ from intron to intron. We therefore introduce several variants of the model, which differ in how much of the data shares a single $\boldsymbol{\beta}$ and $\boldsymbol{\gamma}$.

The gene-specific variant shares a single $\boldsymbol{\beta}$ and a single $\boldsymbol{\gamma}$ across all the introns of a gene. It is still fitted on one gene at a time, and the introns keep their own intercepts and their own terms of the log-likelihood. The estimation, the statistical tests, and the regularization proceed as described above.

The global variant of the model shares $\boldsymbol{\beta}$ and $\boldsymbol{\gamma}$ across all the genes, which are then fitted jointly, so that one $\boldsymbol{\beta}$ and one $\boldsymbol{\gamma}$ describe the whole dataset. We do not regularize this variant, as its LFCs are single values, and there is therefore no collection of the estimates across the genes to fit a prior to. The remaining parameters, that is $\boldsymbol{\alpha}$ and the intercepts, stay specific to the individual genes and introns. The parameter $\boldsymbol{\alpha}$ contains the transcription initiation rate k_{init}, for which a change shared by all the genes is not a plausible description of a biological system. A second and independent reason for not sharing $\boldsymbol{\alpha}$ is discussed in the section Compositional Non-identifiability.

In the gene-specific and the global variant, the estimate is supported by more observations than in the intron-specific one, at the price of assuming that the LFC is common to all the introns or genes that share it. The global variant may also be fitted on a subset of the genes, for example the genes of a single chromosome, which narrows the set over which the LFC is assumed to be common.

A further variant simplifies the model itself, instead of sharing its parameters. As we noted in the section Model Parametrization, the coverage shape of an intron depends on the parameters only through the difference $\boldsymbol{\beta}_i - \boldsymbol{\gamma}_i$ (17c), which is the LFC of the ratio $k_{\text{elong},i}/k_{\text{splice},i}$. We therefore also fit a model in which this difference is a parameter, and whose log-likelihood consists of the coverage term (25) alone. Its only parameters are the difference $\boldsymbol{\beta}_i - \boldsymbol{\gamma}_i$ and the intercepts c_i^π. The exonic and intronic counts are no longer described, so $\boldsymbol{\alpha}$, the intercepts c^{exon} and c_i^{intron}, and the offset terms are not needed. This variant estimates fewer nuisance parameters, and its estimate does not depend on the quantities entering the counts, such as the library sizes or the average transcript lengths. Its LFC may again be specific to an intron, specific to a gene, or shared across all the genes.

3.6 Compositional Non-identifiability

RNA-seq data are compositional (Quinn et al., 2018), meaning that counts describe the proportions of the RNA species within a sample, while the total number of reads of a sample is set by the sequencing run rather than by the amount of RNA in the cells. This is the second source of non-identifiability, in addition to the structural non-identifiability discussed above, and it is a property of the data rather than of our model. In a standard differential expression analysis, a change in the molecule counts shared by all genes is for this reason not identifiable, as multiplying all the counts of a sample by a factor $c \in \mathbb{R}_+$ is absorbed into the library size. Such an analysis therefore estimates an LFC of every gene relative to the other ones, with the reference given by the way in which the library sizes are estimated. The methods, such as the median of ratios (Anders and Huber, 2010), are constructed so that this estimate is not driven by the genes whose counts do change, and the LFCs are then reported relative to the bulk of the genes. We will now discuss what the compositional structure of the data means for our model.

We first note from the steady-state solution (5b)–(5e) that k_{init} scales the abundance of every type of RNA molecule, and that it is the only rate which does so. Multiplying k_{init} of all genes of a sample by a common factor $c \in \mathbb{R}_+$ therefore multiplies the number of RNA molecules of every type in that sample by c, and leaves the proportions between them unchanged. Such a change is absorbed into the library size, exactly as in a standard differential expression analysis, and is not identifiable from RNA-seq data. This is also the second of the two reasons, mentioned in the section Model Variants, why the global variant of our model does not share $\boldsymbol{\alpha}$ across the genes. The parameter $\boldsymbol{\alpha}$ is the LFC of the ratio $k_{\text{init}}/k_{\text{deg}}$ (16a), whose k_{init} part, once shared by all the genes, is exactly the non-identifiable direction described above. A change of k_{deg} shared by all genes can still be captured by the parameters $\boldsymbol{\beta}$ and $\boldsymbol{\gamma}$, as described below.

The intron-specific and the gene-specific variants of our model consequently inherit the assumption of a standard differential expression analysis, which, for a given gene, states that the read counts of the remaining genes, taken as a whole, do not change between the conditions. This global change may come from any of the kinetic rates, for example, from a change of k_{splice} in a large group of genes, which moves their intronic counts. Global changes in kinetic rates will scale all the counts of the gene of interest by a common factor, mimicking a change of k_{init} of that gene. Our model therefore absorbs the violation into $\boldsymbol{\alpha}$, the only parameter which contains k_{init}. Quantities derived from $\boldsymbol{\alpha}$ are also affected by such global changes, for example $\boldsymbol{\alpha} - \boldsymbol{\beta}_i$ (17a), the LFC of the ratio $k_{\text{init}}/k_{\text{elong},i}$. The parameters $\boldsymbol{\beta}_i$ and $\boldsymbol{\gamma}_i$ are not affected, and neither is their difference $\boldsymbol{\beta}_i - \boldsymbol{\gamma}_i$ (17c), which the coverage-only variant of the section Model Variants estimates as a parameter of its own.

The same argument as above also applies when $\boldsymbol{\beta}$ and $\boldsymbol{\gamma}$ are shared across the genes, so the global variant is not affected either. A change of k_{deg}, k_{elong} or k_{splice} shared by all the genes is therefore still captured by $\boldsymbol{\beta}$ and $\boldsymbol{\gamma}$, and k_{init} is the only rate whose globally shared change is non-identifiable.

4 Results

I will apply the model to some data, e.g., ENCODE datasets of stem cell differentiation, or degron-mediated protein KD. Will plot and describe the results here. Should mainly serve for illustration, maybe e.g. mutations known to influence elongation speed (Drosophila and C. Elegans data, maybe human cell lines if data are available).

5 Discussion

Discuss stuff here.

Code availability

An implementation of our method is available at `https://github.com/beyergroup/kinetics-from-total-rna`.

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Supplementary Methods

Implementation details, more math details, etc.

Dynamic Model

- Note about constant k_{elong} across the gene body: This is a somewhat simplifying assumption, as the elongation speed was reported to vary inside a gene body as well, specifically being slower for the first 10-15 kb and then accelerating downstream (Muniz et al., 2021).
- Note about transcript heterogeneity: ODE (exponential decay term) assumes a well-mixed pool of mature transcripts. In some experimental settings, the pool of measured transcripts is not well-mixed: for example, with nascent sequencing via EU-seq, only transcripts finished recently are captured. The mature transcripts in such a pool may not be degraded exponentially. A model that captures the transcript heterogeneity, e.g., with length of polyA tail serving as transcript 'health-bar', may be more appropriate.
- TODO: Add computation of the steady-state solution
- TODO: non-identifiability statement for PDE (to cover the case of our coupled PDE-ODE model).